

02 CV Selection Error Rate vs Sample Size (Polynomial Truth)

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Goal

Data are simulated under a **polynomial** truth. For each sample size on a grid from 50 to 1,000 (step 50), we run 10-fold cross-validation for a **linear** model and a **degree-3 polynomial** model, pick the model with lower mean CV MSE, and record whether CV incorrectly chose the linear model. The **CV selection error rate** is the fraction of 1,000 Monte Carlo replications where linear wins.

Simulation settings

Table 1: Simulation and CV settings for analysis 02 (polynomial truth).

setting	value
seed	2026
K	10
Monte Carlo replications	1000
per n	
Sample sizes (grid)	50, 100, ..., 1000 (step 50)
beta0	1
beta1	2
beta2	1
beta3	1
sigma	4
poly_degree	3
truth	polynomial: $y = \text{beta0} + \text{beta1}x + \text{beta2}x^2 + \text{beta3}x^3 + \text{eps}$

setting	value
selection rule	Choose poly3 if mean CV MSE strictly lower than linear; else linear

CV error summary by sample size

Table 2: Monte Carlo mean selection-error rate (truth is polynomial; error = linear chosen) and approximate MC standard error.

n	error_rate	mc_se
50	0.192	0.01246
100	0.013	0.00358
150	0.002	0.00141
200	0.000	0.00000
250	0.000	0.00000
300	0.000	0.00000
350	0.000	0.00000
400	0.000	0.00000
450	0.000	0.00000
500	0.000	0.00000
550	0.000	0.00000
600	0.000	0.00000
650	0.000	0.00000
700	0.000	0.00000
750	0.000	0.00000
800	0.000	0.00000
850	0.000	0.00000
900	0.000	0.00000
950	0.000	0.00000
1000	0.000	0.00000

CV error rate vs sample size

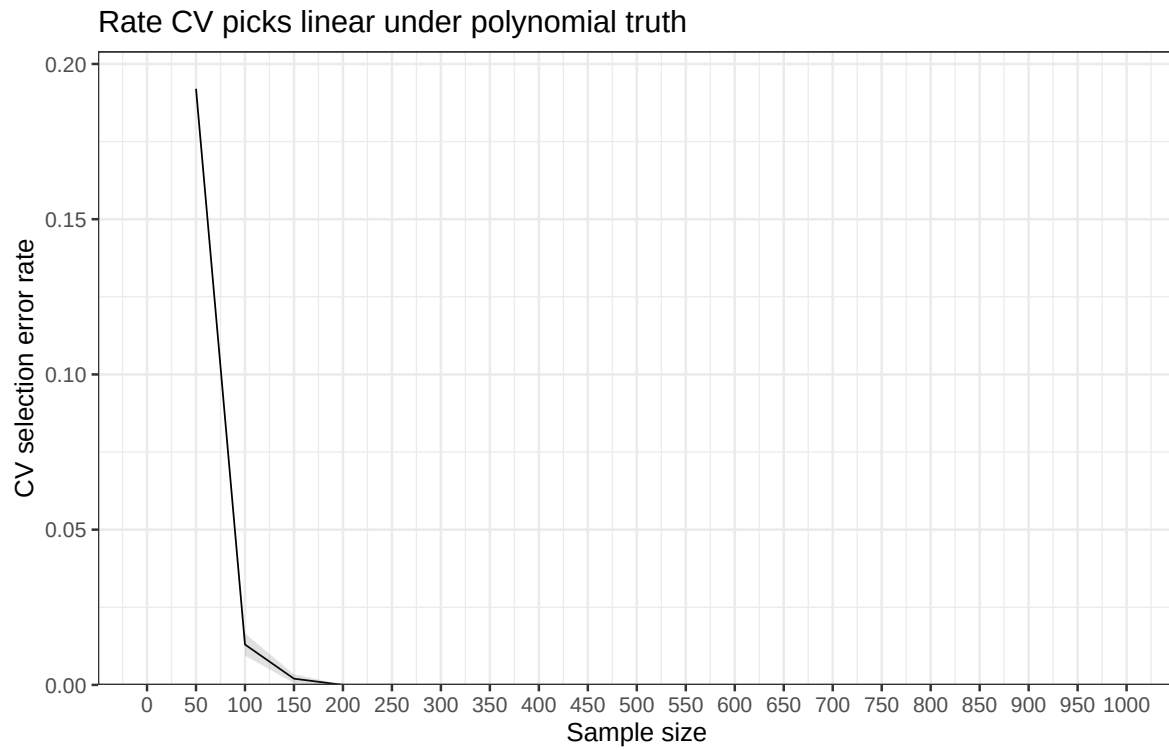


Figure 1: CV selection error rate as a function of sample size (ribbon = MC SE band, clipped to $[0, 1]$).